

Manyuan Zhang

Personal Data

Position	Staff Researcher at Meituan LongCat, Hong Kong
Research	Multimodal foundation models, generative models, and physical intelligence
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Education

08/2021–07/2025	Ph.D., Multimedia Laboratory, The Chinese University of Hong Kong Advisors: Prof. Xiaogang Wang and Prof. Hongsheng Li.
09/2015–06/2019	B.Eng., Network Engineering, University of Electronic Science and Technology of China GPA: 3.90/4.00.

Work Experience

05/2025–Present	Meituan LongCat, Staff Researcher - Selected for the highest tier of Meituan’s Beidou Program. First foundation-model researcher hired specifically for Meituan’s Hong Kong site. - Contributed to the flagship LongCat-Flash, a 560B-parameter MoE language model. - Lead native multimodal generation for LongCat-Next. Manage a team of approximately 10 researchers and interns, build petabyte- and billion-sample-scale data pipelines, and own the end-to-end architecture design and training of the generation stack. - Serve as training lead for Meituan Robotics’ first VLA and native embodied models. Achieved leading open-source results on major simulation benchmarks and consistently outperformed $\pi_{0.5}$ in real-robot deployment.
02/2019–05/2025	SenseTime Research Foundation Model Group, Researcher / Research Intern - Developed frontier vision models as first author or a core contributor. Delivered winning systems for the CVPR Masked Face Recognition Challenge, ICCV video face recognition challenges, and U.S. NIST FRVT, as well as the CVPR ActivityNet and Open Images challenges in video recognition and object detection. - Conducted foundation-model and AIGC research and built the SenseMirage image-generation model and product from scratch. SenseMirage exceeded one million daily active users within a month of launch, a first in SenseTime’s history, and received a special award from the CEO. - Member of the Foundation Model Group when it received SenseTime’s highest company honor in both 2022 and 2023. Group leaders included Yu Liu, founder of Vivix, and Junjie Yan, founder of MiniMax.
05/2018–01/2019	Research Intern, Megvii (Face++) / ByteDance AI Lab Research on face recognition.

Selected Competition Awards

- 1st place**, CVPR 2021 Masked Face Recognition Challenge: WebFace260M and both InsightFace tracks.
- 1st place**, CVPR 2020 ActivityNet Challenge: Kinetics-700 and AVA tracks.
- 1st place**, U.S. NIST FRVT 1:1 Verification and 1:N Identification.
- 1st place**, ICCV 2019 Multi-Moments in Time Video Recognition Challenge.
- 1st place**, ICCV 2019 Lightweight Face Recognition Challenge.

Selected Research and Engineering Projects

2026–Present	Meituan Robotics 0/0.5 - Responsible for training the first VLA system, including the framework and the complete pretraining, mid-training, and post-training stack. Jointly train a VLM + MMDiT architecture on tens of thousands of hours of cross-embodiment data and VQA tasks. - Meituan Robotics 0 achieves leading results among open-source VLA models on RoboTwin 2.0 and RoboDojo, including first place in open, zero-shot settings, and consistently outperforms $\pi_{0.5}$ in real-robot deployment. - Built a native embodied model with a three-expert mixture-of-transformers architecture that unifies understanding, video generation, and action prediction in an end-to-end loop.
2025–2026	LongCat-Next - Lead native multimodal generation and manage a team of approximately 10 researchers and interns. Own the generation architecture, data systems, training, and evaluation lifecycle. - Developed a unified autoregressive model that represents text, vision, and audio as discrete tokens using DiNA and a dNaViT visual tokenizer. - Built petabyte- and billion-sample-scale pipelines for cleaning, recaptioning, rebalancing, filtering, and expert distillation; led multi-stage training and the human-evaluation Arena.
2025–2026	LLaVA-OneVision-2.0 and OneVision-Encoder - Trained a 600M-parameter vision transformer using billion-scale image feature clustering and million-scale semantic concept prototypes for structured visual representation learning. - The encoder outperformed Qwen3-ViT and SigLIP2 across 16 image, video, and document benchmarks. On Diving48 attentive probing, it improved Top-1 accuracy by 17.1 and 8.1 points over SigLIP2 and DINOv3, respectively.
2022–2025	SenseMirage - Built a text-to-image platform from scratch. It surpassed one million daily active users after launch, served more than 100 enterprise customers, and handled over 100,000 API calls per day. - Developed a pipeline for billions of raw images covering crawling, OCR, watermark removal, bucketing, captioning, and quality control. - Built human-preference quality models for filtering, loss weighting, DPO, and evaluation, as well as a LoRA architecture-search pipeline for commercial customization.
2019–2021	SenseTime Vision Model AutoML Framework - Built an end-to-end AutoML framework that jointly optimized data cleaning, architecture, loss design, and on-device inference cost, and used it to study scaling laws for vision models. - Deployed the framework across more than 50 product lines. It powered winning NIST FRVT 1:1 and 1:N systems and enabled rapid development of the CVPR 2021 Masked Face Recognition Challenge winner.

Selected Publications

More than 30 papers at leading conferences; 2,000+ Google Scholar citations. [†]Project lead or corresponding author. ^{*}First author.

- UniClawBench: A Universal Benchmark for Proactive Agents on Real-World Tasks**
Zhekai Chen, Chengqi Duan, Kaiyue Sun, Bohao Li, Yuqing Wang, **Manyuan Zhang**[†], Xihui Liu
arXiv, 2026
- LLaVA-OneVision-2.0: Towards Next-Generation Perceptual Intelligence**
LMMs-Lab, including **Manyuan Zhang** (published as Nansen Zhang)
Technical Report, 2026 [机器之心]

- **OneVision-Encoder: Codec-Aligned Sparsity as a Foundational Principle for Multimodal Intelligence**
LMMs-Lab, including **Manyuan Zhang**
Technical Report, 2026
- **Gen-Searcher: Reinforcing Agentic Search for Image Generation**
Kaituo Feng, **Manyuan Zhang**[†], Shuang Chen, et al.
arXiv, 2026 [机器之心]
- **Think with 3D: Geometric Imagination Grounded Spatial Reasoning from Limited Views**
Zhangquan Chen, **Manyuan Zhang**[†], Xinlei Yu, et al.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026 [机器之心]
- **OneThinker: All-in-One Reasoning Model for Image and Video**
Kaituo Feng, **Manyuan Zhang**[†], Hongyu Li, et al.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026 [量子位]
- **EditThinker: Unlocking Iterative Reasoning for Any Image Editor**
Hongyu Li, **Manyuan Zhang**[†], Dian Zheng, et al.
arXiv, 2025
- **CodePlot-CoT: Mathematical Visual Reasoning by Thinking with Code-Driven Images**
Chengqi Duan, Kaiyue Sun, Rongyao Fang, **Manyuan Zhang**[†], et al.
CVPR Findings, 2026
- **Thinking-while-Generating: Interleaving Textual Reasoning throughout Visual Generation**
Ziyu Guo, Renrui Zhang, Hongyu Li, **Manyuan Zhang**[†], et al.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026 [量子位]
- **CTR3D: Cross-View Token Reduction for Dense Multi-View Generation**
Kunming Luo, Hongyu Yan, Yuan Liu, Zihao Zhang, **Manyuan Zhang**, et al.
International Conference on 3D Vision (3DV), 2026; Best Paper Award Nomination
- **Exploring Reasoning Reward Model for Agents**
Kaixuan Fan, Kaituo Feng, **Manyuan Zhang**[†], et al.
Annual Meeting of the Association for Computational Linguistics (ACL), 2026 [机器之心]
- **Lumina-Image 2.0: A Unified and Efficient Image Generative Framework**
Qi Qin, Le Zhuo, Yi Xin, et al., **Manyuan Zhang**, et al.
International Conference on Computer Vision (ICCV), 2025
- **Let's Verify and Reinforce Image Generation Step by Step**
Renrui Zhang, Chengzhuo Tong, Zhizheng Zhao, et al., **Manyuan Zhang**, et al.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025 [机器之心]
- **Three Things We Need to Know About Transferring Stable Diffusion to Visual Dense Prediction Tasks**
Manyuan Zhang^{*}, Guanglu Song, Yu Liu, Hongsheng Li
European Conference on Computer Vision (ECCV), 2024
- **Decoupled DETR: Spatially Disentangling Localization and Classification for Improved End-to-End Object Detection**
Manyuan Zhang^{*}, Guanglu Song, Yu Liu, Hongsheng Li
International Conference on Computer Vision (ICCV), 2023
- **Towards Robust Face Recognition with Comprehensive Search**
Manyuan Zhang^{*}, Guanglu Song, Yu Liu, Hongsheng Li
European Conference on Computer Vision (ECCV), 2022
- **Discriminability Distillation in Group Representation Learning**
Manyuan Zhang^{*}, Guanglu Song, Hang Zhou, Yu Liu
European Conference on Computer Vision (ECCV), 2020

Open-Source Projects

- **X-Temporal** (500+ stars) github.com/Sense-X/X-Temporal
Distributed PyTorch framework for video-understanding research. Lead contributor.
- **DI-drive** (500+ stars) github.com/opencv/DI-drive
Decision-intelligence platform for autonomous-driving simulation. Led the imitation-learning module.
- **DI-star** (1,200+ stars) github.com/opencv/DI-star
Distributed StarCraft II training platform. Built the supervised-learning component and achieved 6,000+ MMR with Zerg.